

Clustering and Similarity

Lecturer Prof. Yu Li Scribe _____

Clustering organizes objects into groups using a chosen representation and a measure of resemblance. This part of the lecture explains why grouping is useful, how objects become numerical data, and how similarity or distance can be calculated. The central question is what “alike” should mean for the task.

1 Why form clusters

Lecture connection: PDF pp. 56–59.

A cluster contains objects that are relatively similar to one another and relatively different from objects in other groups. The slide describes small intra-cluster differences and large inter-cluster differences. This is a guiding objective; its meaning depends on the features and comparison measure.

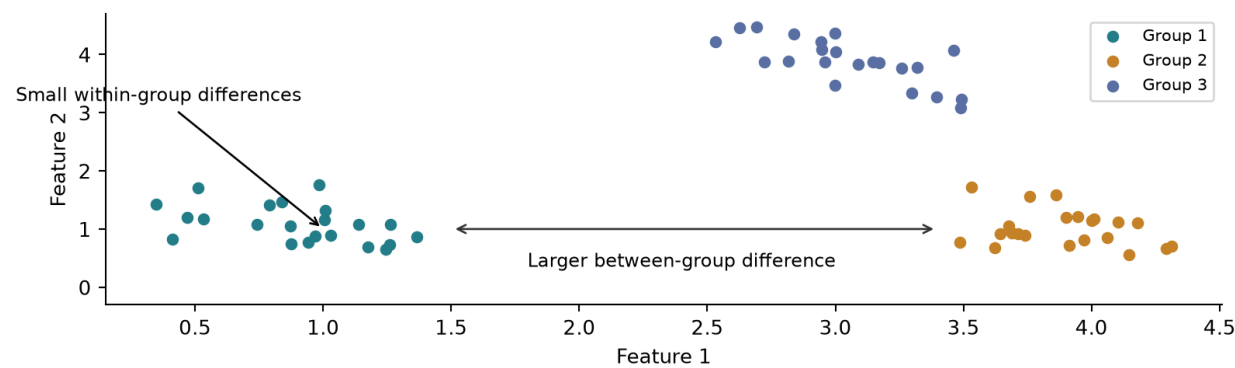


Figure 1. Synthetic example of compact, separated groups. The colors illustrate a grouping, not the output of a particular clustering algorithm.

1.1 Organization and biological discovery

Setting	Purpose of grouping
Items or documents	Organize related items and make browsing or searching easier.
People	Find groups with related needs; groups need not follow age or gender.
Genes	Explore co-expression patterns and possible shared biological functions.
Samples or cells	Explore disease subtypes or potential cell types.

Co-expressed genes can suggest shared regulation or pathways, but similarity alone does not establish function. The slides also connect clustering to disease-related expression. A cluster pattern can motivate a differential-expression question; a statistical test is still needed to establish a difference between defined groups.

Clustering may be a stand-alone exploration tool, a preprocessing step, or a way to summarize a large dataset. The slides mention privacy as a motivation for summaries. Grouping alone does not guarantee anonymity or remove disclosure risk.

2 From objects to cluster assignments

Lecture connection: PDF pp. 60–68.

2.1 The fruit example

People readily group apples, bananas and oranges in pictures. A computer receives numerical representations of those pictures. The slide sequence moves from images to numerical data and then to a clustering method that produces an indicator for each item.

Representation is a substantive choice. Pixel values describe an image directly, while features can emphasize properties such as color or shape. These are examples of possible representations, not a feature-extraction method specified by the slides. A change in background or appearance can affect numerical similarity even when the depicted fruit is the same.

Item IDs in the slide	Cluster indicator	Visual grouping
F1, F3, F8	1	Bananas
F2, F4, F6	2	Apples or apple-like imagery
F5, F7, F9	3	Oranges

The indicator numbers name groups; they are not scores or ranks. Relabeling groups 1, 2 and 3 as 3, 1 and 2 preserves the same partition. The meaningful result is which objects share a group.

2.2 Three required ingredients

Ingredient	Question to answer
Data representation	What is one object, and which features describe it?
Similarity or dissimilarity	Which differences should matter, and how much?
Clustering algorithm	What procedure forms groups from the data and measure?

This mirrors the earlier sequence-comparison setup: sequences, a scoring rule, and an alignment procedure. Choosing a comparison measure and choosing an algorithm are separate decisions. These pages introduce the setup rather than a detailed clustering algorithm.

2.3 Check the interpretation

Inspect whether within-group objects are similar and whether groups differ under the chosen representation. In the fruit example, visual categories provide an intuitive check. With biological data, an apparent cluster should be compared with biological context and technical factors; a pattern may reflect preprocessing or measurement conditions.

For the gene-expression project on p. 68, collect and merge data if needed, explore it, visualize it and clean it. Clustering is the proposed next stage, while dimensionality reduction is introduced later in the course. Preserve row and column meanings and document preprocessing decisions.

3 Similarity and cosine comparison

Lecture connection: PDF pp. 71–72.

Similarity increases as objects become more alike. Dissimilarity decreases as they become more alike, often reaching zero for identical objects. Similarities often lie in $[0,1]$, but that range is not universal: cosine similarity and correlation can be negative. Distances may have no fixed upper bound.

3.1 Compare vector direction

For nonzero vectors x and y , cosine similarity is their dot product divided by the product of their lengths:

$$\cos(x, y) = \sum x_k y_k / [\sqrt{(\sum x_k^2)} \sqrt{(\sum y_k^2)}]$$

Cosine measures direction rather than overall magnitude. A positive multiple of a vector has cosine similarity 1 with the original. Orthogonal vectors have cosine 0; opposite directions have cosine -1 . For nonnegative vectors the range is $[0,1]$. A zero vector makes the formula undefined.

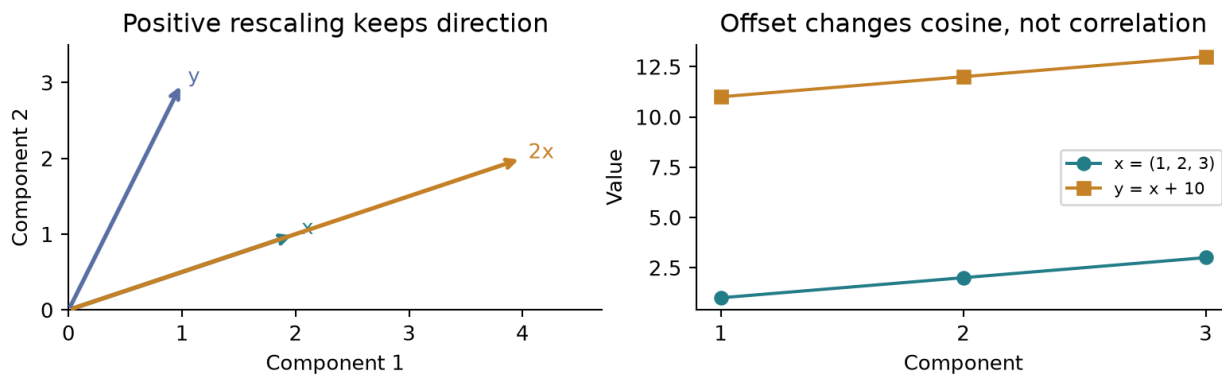


Figure 2. Original illustrations of direction and offsets. Left: x and $2x$ point in the same direction. Right: adding a constant preserves a linear profile but changes its direction from the origin.

3.2 Recalculate the slide example

$x = (3, 2, 0, 5, 0, 0, 0, 2, 0, 0)$

$y = (1, 0, 0, 0, 0, 0, 0, 1, 0, 2)$

Quantity	Calculation
Dot product	$3 \times 1 + 2 \times 1 = 5$
Squared lengths	$ x ^2 = 9 + 4 + 25 + 4 = 42$; $ y ^2 = 1 + 1 + 4 = 6$
Cosine similarity	$5 / (\sqrt{42} \times \sqrt{6}) = 5 / \sqrt{252} \approx 0.3150$

Numerical correction: PDF p. 72 prints $\sqrt{6}$ as 2.245. Its correct value is about 2.449; the slide's final cosine value of 0.3150 is consistent with the correct square root.

Cosine can be useful when proportional feature profiles should count as similar. It is not invariant to adding the same constant to every component. Centered profiles are compared by Pearson correlation, discussed next.

4 Correlation measures linear association

Lecture connection: PDF pp. 73–74.

4.1 Center before comparing

Pearson correlation measures the direction and strength of a linear relationship between paired values. Subtract each variable's mean, then normalize the dot product of the centered values:

$$r = \Sigma[(x_i - \bar{x})(y_i - \bar{y})] / \sqrt{\{\Sigma(x_i - \bar{x})^2 \times \Sigma(y_i - \bar{y})^2\}}$$

This is also covariance divided by the product of standard deviations when the conventions are consistent. Correlation is cosine similarity applied to mean-centered vectors. Its value is undefined if either variable is constant.

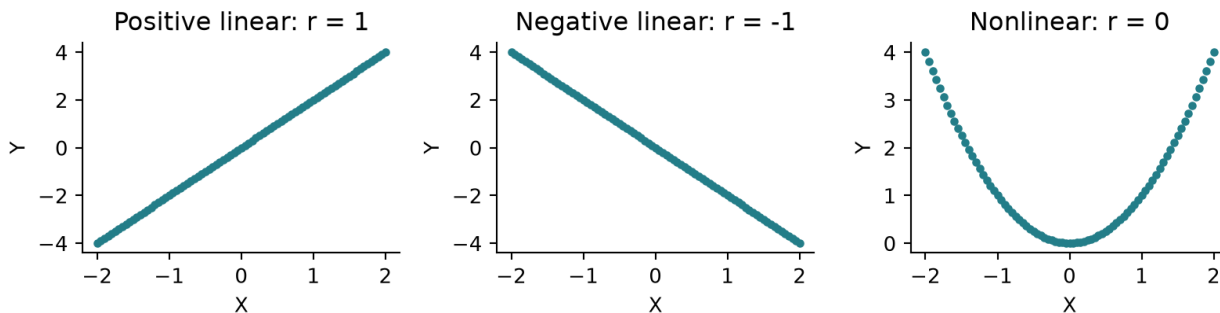


Figure 3. Synthetic examples of positive, negative and nonlinear relationships. The symmetric quadratic example has zero linear correlation despite a clear relationship.

$r = 1$ means a perfect increasing linear relation; $r = -1$ means a perfect decreasing linear relation. Near-zero correlation means little linear association, not necessarily no relationship. Correlation also does not establish causation. The horizontal constant case shown among the slide illustrations should be treated as undefined, not as a valid $r = 1$ example.

4.2 Temperature and sales calculation

The slide example uses 12 paired temperature and sales observations. Calculate both means, form deviations $a = \text{temperature} - \text{mean temperature}$ and $b = \text{sales} - \text{mean sales}$, then sum ab , a^2 and b^2 . Using the rounded totals printed on the slide:

Mean temperature	Mean sales	Σab	Σa^2	Σb^2
18.7 °C	\$402	5,325	177.0	174,757

$r \approx 5,325 / \sqrt{(177.0 \times 174,757)} \approx 0.95745$ (the slide reports 0.9575). This is a strong positive linear association in the example. The displayed means and intermediate values are rounded; recalculation from unrounded inputs may differ slightly.

4.3 Cosine and correlation answer different questions

For $x = (1, 2, 3)$ and $y = (11, 12, 13)$, Pearson $r = 1$ because $y = x + 10$. Cosine similarity is $74 / \sqrt{6076} \approx 0.9493$, since the offset changes direction from the origin. Euclidean distance is $\sqrt{300} \approx 17.32$. Thus identical linear patterns can coexist with different absolute levels.

5 Euclidean distance and a distance matrix

Lecture connection: PDF pp. 75–76.

Euclidean distance is straight-line separation in numerical feature space. For m attributes:

$$d_2(p, q) = \sqrt{[\sum_{k=1}^m (p_k - q_k)^2]}$$

Each coordinate difference is squared, the squared differences are added, and the square root restores the original scale when coordinates share a unit. Larger differences contribute more strongly because of the squaring.

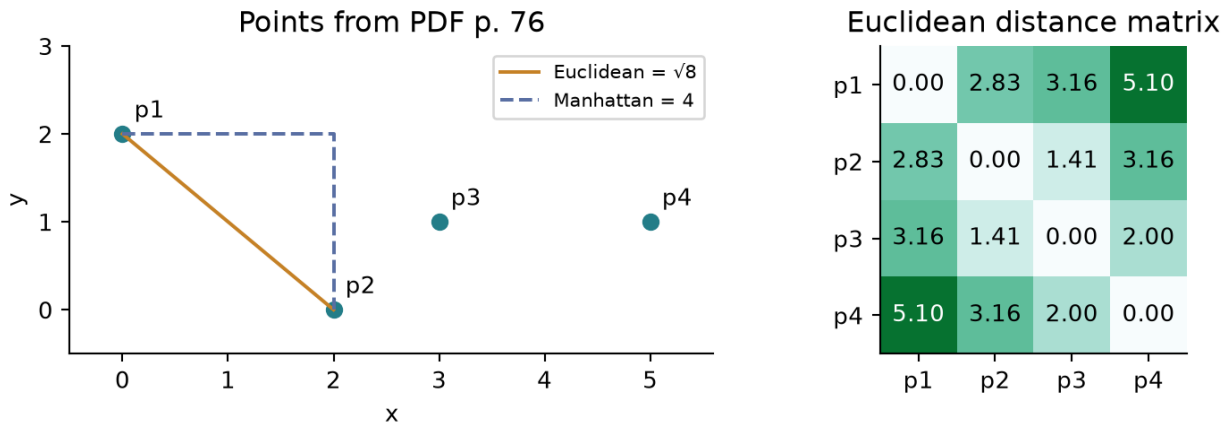


Figure 4. Redrawn geometry and calculated Euclidean distances for $p1 = (0,2)$, $p2 = (2,0)$, $p3 = (3,1)$, $p4 = (5,1)$. Values match PDF p. 76 up to rounding.

5.1 Work through one pair

$$d_2(p1, p2) = \sqrt{[(0 - 2)^2 + (2 - 0)^2]} = \sqrt{8} \approx 2.828.$$

$$d_2(p2, p3) = \sqrt{[(2 - 3)^2 + (0 - 1)^2]} = \sqrt{2} \approx 1.414.$$

	p1	p2	p3	p4
p1	0	2.828	3.162	5.099
p2	2.828	0	1.414	3.162
p3	3.162	1.414	0	2.000
p4	5.099	3.162	2.000	0

A pairwise distance matrix is symmetric and has zeros on its diagonal. The smallest off-diagonal entry identifies $p2$ and $p3$ as the closest pair under Euclidean distance. It describes pairwise separation; it is not itself a set of cluster assignments.

5.2 Scaling changes what the distance emphasizes

For the height–weight example, the differences between $P1$ and $P2$ are 0.15 m and 21 kg, so an unscaled distance is $\sqrt{(0.15^2 + 21^2)} \approx 21.0005$. After the slide’s min–max scaling, the same two records are (0.625, 0.875) and (0, 0), giving distance about 1.0753.

The transformed distance answers a different comparison question: both observed feature ranges receive comparable scale. Scaling should follow the intended role of each attribute; units and feature weighting should not be left accidental.

6 Minkowski distances and choosing a measure

Lecture connection: PDF pp. 77–80.

Minkowski distance generalizes coordinate-based distances using a parameter r :

$$d_r(p,q) = [\sum_{k=1}^m |p_k - q_k|^r]^{1/r}, \text{ with } r \geq 1 \text{ for a metric.}$$

Parameter	Name	Rule
$r = 1$	Manhattan or L1	Add absolute coordinate differences.
$r = 2$	Euclidean or L2	Square, sum and take a square root.
$r \rightarrow \infty$	Maximum or L_∞	Take the largest absolute coordinate difference.

Points at distance 1 from the origin

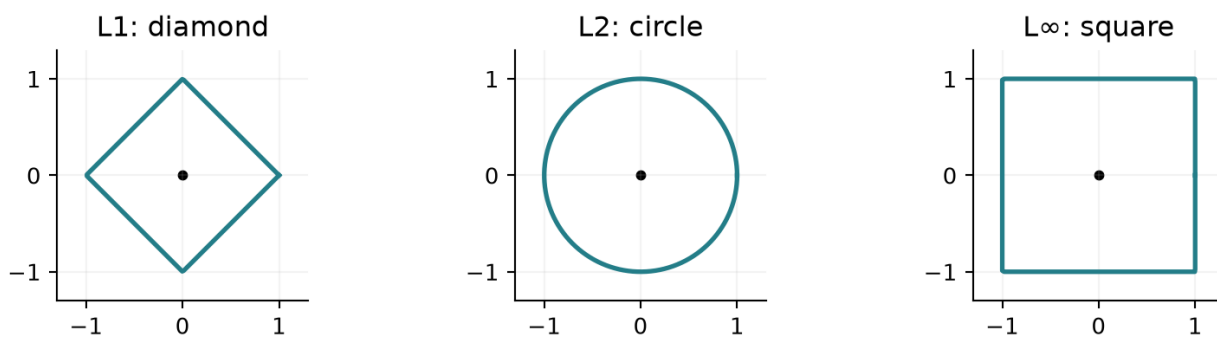


Figure 5. Original geometric comparison. Different distance rules give different sets of points at unit distance from the origin.

6.1 Compare the same pairs under different rules

Pair	Absolute differences	L1	L2	L_∞
p1,p2	(2,2)	4	2.828	2
p1,p3	(3,1)	4	3.162	3
p1,p4	(5,1)	6	5.099	5
p2,p3	(1,1)	2	1.414	1
p2,p4	(3,1)	4	3.162	3
p3,p4	(2,0)	2	2.000	2

These pairs reconstruct the matrices on p. 79. L1 ties p1–p2 and p1–p3, while L2 and L_∞ distinguish them. Nearest-neighbor relations and later grouping can therefore depend on the chosen measure.

For binary vectors, L1 equals Hamming distance: each different bit contributes 1. Example: (1,0,1,1) versus (0,0,1,0) differs in two positions, so both distances equal 2.

Use cosine for direction, correlation for centered linear patterns, and coordinate distances for absolute differences in appropriately scaled features. Document the choice together with the representation and preprocessing.

Source: BMEG3105 Lecture 5, supplied PDF, physical pp. 56–80. Figures are prepared with OpenAI Codex assistance.